Equalities, Inequalities and Minimizers

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Equalities

Mastery of Elementary Algebra (a.k.a. "College Algebra") is a prerequisite for calculus, which is in turn a prerequisite for most statistics and data science courses. Nevertheless, each year, some students are still uncomfortable with algebraic manipulations of mathematical formulas. Therefore, I include this section as a quick reference.

Definition 0.1 (Transitive relation)**.** Let $\sim$ be a relation on a set S. The relation $\sim$ is **transitive** if, for all $a, b, c \in S$, $a \sim b$ and $b \sim c$ together imply $a \sim c$.

Example 0.1 (A transitive relation and one that is not)**.**

- $\leq$ is transitive: for example, $1 \leq 2$ and $2 \leq 5$, and indeed $1 \leq 5$.
- $\neq$ is not transitive: $1 \neq 2$ and $2 \neq 1$, but $1 \neq 1$ is false.

Theorem 0.1 (Equalities are transitive)**.** *If $a = b$ and $b = c$, then $a = c$. That is, $=$ is a transitive relation.*

Theorem 0.2 (Substituting equivalent expressions)**.** *If $a = b$, then for any function $f(x)$, $f(a) = f(b)$*

Definition 0.2 (Identity)**.** An **identity** is an equation between two expressions that holds for every value of its variables in a stated set, such as the real numbers.

Example 0.2 (An identity, and an equation that is not one)**.**

- $2(x + 1) = 2x + 2$ is an identity: it holds for every real x. For example, at $x = 3$, both sides equal 8.

- $x + 1 = 3$ is not an identity: it holds at $x = 2$, but at $x = 0$ the left side is 1 and the right side is 3.

Inequalities

Definition 0.3 (Strict inequalities)**.** For real numbers a and b, $a < b$ ("a is less than b") and $a > b$ ("a is greater than b") are **strict inequalities**.

Definition 0.4 (Non-strict inequalities)**.** For real numbers a and b, the **non-strict inequalities** also allow equality, unlike the strict inequalities (Definition 0.3):

- $a \leq b$ means $a < b$ or $a = b$;
- $a \geq b$ means $a > b$ or $a = b$.

Example 0.3 (Strict and non-strict inequalities)**.**

- $3 \leq 3$ is true, because $3 = 3$, but $3 < 3$ is false.
- $2 < 3$ and $2 \leq 3$ are both true.
- $4 \geq 5$ is false, because $4 > 5$ and $4 = 5$ are both false.

Theorem 0.3 (Adding to both sides of an inequality)**.** *If $a < b$, then $a + c < b + c$*

Theorem 0.4 (Negating both sides of an inequality)**.** *If $a < b$, then: $-a > -b$*

Theorem 0.5 (Multiplying both sides of an inequality by a positive number)**.** *If $a < b$ and $c > 0$, then $ca < cb$.*

Theorem 0.6 (Negation is multiplication by -1)**.**

$$-a = (-1) * a$$

Powers, square roots, and absolute values

Definition 0.5 (Power)**.** Let a be a real number and n a natural number. The nth **power** of a is the product of n copies of a:

$$a^n \stackrel{\text{def}}{=} \underbrace{a \cdot a \cdot \cdots \cdot a}_{n \text{ factors}}$$

Definition 0.6 (Base of a power)**.** In a^n (Definition 0.5), a is the **base**.

Definition 0.7 (Exponent of a power)**.** In a^n (Definition 0.5), n is the **exponent**.

Definition 0.8 (Square)**.** The second power a^2 (Definition 0.5) is the **square** of a.

Example 0.4 (Powers)**.**

- $$\begin{aligned} 2^3 &= 2 \cdot 2 \cdot 2 \\ &= 8, \end{aligned}$$

  with base 2 and exponent 3.

- $$\begin{aligned} (-3)^2 &= (-3) \cdot (-3) \\ &= 9, \end{aligned}$$

  but

  $$\begin{aligned} -3^2 &= -(3^2) \\ &= -9 : \end{aligned}$$

  the exponent applies only to the 3.

- $5^0 = 1$.

- $$\begin{aligned} 2^{-2} &= \frac{1}{2^2} \\ &= \frac{1}{4}. \end{aligned}$$

Definition 0.9 (Powers with exponent zero or a negative integer)**.** For the exponent 0 and negative integer exponents:

- $a^0 \stackrel{\text{def}}{=} 1$, including $0^0 = 1$ by convention;
- $a^{-n} \stackrel{\text{def}}{=} \frac{1}{a^n}$, when $a \neq 0$.

Remark (Powers with real exponents)

Remark 0.1 (Powers with real exponents). Definition 21 in Polynomials, Exponentials and Logarithms extends powers to real exponents when the base is positive.

Definition 0.10 (Square root)**.** For a real number $a \geq 0$, the **square root** of a, written $\sqrt{a}$, is the non-negative real number whose square (Definition 0.5) is a.

Example 0.5 (Square roots)**.**

- $\sqrt{9} = 3$. Although $(-3)^2 = 9$ too, -3 is negative, so $\sqrt{9}$ is not -3.
- $\sqrt{0} = 0$.
- $\sqrt{2} \approx 1.41421$.
- $\sqrt{-4}$ is not a real number, because no real number squares to -4 (see Section 1 in Complex Numbers).

Definition 0.11 (Absolute value of a real number)**.** The **absolute value** of a real number a is

$$|a| \stackrel{\text{def}}{=} \begin{cases} a, & a \geq 0 \\ -a, & a < 0 \end{cases}$$

Example 0.6 (Absolute values)**.**

- $|2.5| = 2.5$, because $2.5 \geq 0$.
- $$\begin{aligned} |-3| &= -(-3) \\ &= 3, \end{aligned}$$

  because $-3 < 0$.
- $|0| = 0$.

- $$\begin{aligned} |-3| &= 3 \\ &= \sqrt{9} \\ &= \sqrt{(-3)^2}: \end{aligned}$$

  for every real a, $|a| = \sqrt{a^2}$ (Definition 0.10).

- $|x-1| < 1$ says that x is less than 1 away from 1, that is, $0 < x < 2$.

Remark (Absolute value as distance)

Remark 0.2 (Absolute value as distance). For real numbers a and b, $|a-b|$ is the distance between a and b on the number line.

Minimum, maximum, argmin and argmax

Exercise 0.1 (Smallest value, and where it occurs)**.**

1. Let $f(x) = (x-2)^2 + 1$ for every $x \in \mathbb{R}$. What is the smallest value f takes, and at which x does it take that value?
2. Now let $g(x) = (x-2)^2 + 1$ only for $x \in \{0, 1, 3, 4\}$. What is the smallest value g takes, and at which x does it take that value?

Solution

Solution 0.1.

1. The square of a real number is never negative, so $(x-2)^2 \geq 0$ for every x, and therefore $f(x) \geq 0 + 1 = 1$. The square is 0 only when $x - 2 = 0$, that is, when $x = 2$, and there $f(2) = 1$. So the smallest value of f is 1, and $x = 2$ is the only input where f takes it.

2. Evaluate g at each of its four inputs:

x	0	1	3	4
$g(x)$	5	2	2	5

The smallest value of g is 2, and g takes it at two inputs, $x = 1$ and $x = 3$. The answer to "where?" is the set $\{1, 3\}$, not a single number.

Definition 0.12 (Minimum)**.** Let $A \subseteq \mathbb{R}$. A number m is the **minimum** of A, written $\min A$, if $m \in A$ and $m \leq a$ for all $a \in A$.

Remark (Not every set has a minimum)

Remark 0.3 (Not every set has a minimum). The set $\{3, 1, 4\}$ has minimum 1, and the interval $[0, 1]$ has minimum 0. The interval $(0, 1]$ has no minimum: every element a of it has a smaller element, $a/2$, also in it.

Definition 0.13 (Maximum)**.** Let $A \subseteq \mathbb{R}$. A number M is the **maximum** of A, written $\max A$, if $M \in A$ and $a \leq M$ for all $a \in A$.

Example 0.7 (A maximum, and a set without one)**.**

- $\max\{-2, 5, 0\} = 5$: 5 is in the set, and $-2 \leq 5$, $5 \leq 5$, $0 \leq 5$.
- The interval $[0, 1)$ has no maximum. For any $a \in [0, 1)$, the number $\frac{a+1}{2}$ is also in $[0, 1)$, and it is larger than a, since $\frac{a+1}{2} > a$ exactly when $a < 1$. So no element of $[0, 1)$ is at least as large as every element.

Definition 0.14 (Argmin)**.** Let $f : A \to \mathbb{R}$ be a function. The **argmin** of f over A is the set of inputs where f takes its smallest value:

$$\arg\min_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A,\ f(x) \leq f(a)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg\min_{x \in A} f(x)$.

Remark (Smallest value versus where it occurs)

Remark 0.4 (Smallest value versus where it occurs). The smallest value itself is $\min f(A)$, the minimum (Definition 0.12) of the image of f, and the argmin is where that value is attained. For example, let $f(x) = x^2$ on $A = \{-1, 0, 2\}$. The image is $f(A) = \{1, 0, 4\}$, so the smallest value is $\min f(A) = 0$, and $\arg\min_{x \in A} f(x) = \{0\}$.
The argmin is empty when f has no smallest value, for example $f(x) = x$ on $(0, 1]$.

Definition 0.15 (Argmax)**.** Let $f : A \to \mathbb{R}$ be a function. The **argmax** of f over A is the set of inputs where f takes its largest value:

$$\arg\max_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A,\ f(a) \leq f(x)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg\max_{x \in A} f(x)$.

Example 0.8 (An argmax with two points, and an empty one)**.**

- Let $g(x) = x^2$ on $A = \{-2, 0, 1, 2\}$. The values are $g(-2) = 4$, $g(0) = 0$, $g(1) = 1$ and $g(2) = 4$, so the largest value is 4 and $\arg\max_{x \in A} g(x) = \{-2, 2\}$. This argmax has two elements, so it is not written as a single $\hat{x}$.
- Let $f(x) = x$ on $[0, 1)$. As in Example 0.7, for every input x the input $\frac{x+1}{2}$ has a larger value, so no input attains a largest value, and $\arg\max_{x \in [0,1)} f(x)$ is the empty set.

Global and local minimizers

Exercise 0.2 (A dip that is not the bottom)**.** Let $f(x) = x^3 - 3x$ for $x \in \mathbb{R}$.

1. Factor $f(x) - f(1)$ completely.
2. Use part 1 to compare $f(x)$ with $f(1)$ for every x in the interval $(0, 2)$. Is $f(x)$ larger than, smaller than, or equal to $f(1)$, and for which x?
3. Compute $f(-3)$. Is $f(1)$ the smallest value f takes on all of $\mathbb{R}$?
4. Does f take a smallest value anywhere on $\mathbb{R}$?

Solution

Solution 0.2.

1. Since

$$\begin{aligned} f(1) &= 1 - 3 \\ &= -2, \end{aligned}$$

so

$$\begin{aligned} f(x) - f(1) &= x^3 - 3x + 2 \\ &= (x-1)(x^2 + x - 2) \\ &= (x-1)(x-1)(x+2) \\ &= (x-1)^2(x+2). \end{aligned}$$

2. For $x \in (0,2)$, $(x-1)^2 \geq 0$ and $x+2 > 0$, so their product is at least 0. By part 1, $f(x) - f(1) \geq 0$, that is, $f(x) \geq f(1)$. So $f(x)$ is never smaller than $f(1)$ on $(0,2)$. It equals $f(1)$ only when $(x-1)^2 = 0$, that is, only at $x = 1$, and it is larger than $f(1)$ at every other $x \in (0,2)$.

3. $$\begin{aligned} f(-3) &= -27 + 9 \\ &= -18. \end{aligned}$$

   Since $-18 < -2 = f(1)$, $f(1)$ is not the smallest value of f on $\mathbb{R}$.

4. No. For each $n \geq 2$,

   $$\begin{aligned} f(-n) &= -n^3 + 3n \\ &= -n(n^2 - 3) \\ &\leq -n, \end{aligned}$$

   so no value of f is smaller than all the others: for any x, picking $n \geq 2$ with $-n < f(x)$ gives $f(-n) < f(x)$.

Definition 0.16 (Global minimizer)**.** Let $A \subseteq \mathbb{R}^p$ and let $f : A \to \mathbb{R}$ be a function. A point $x^* \in A$ is a **global minimizer** of f over A if $f(x^*) \leq f(x)$ for all $x \in A$.

Remark (Global minimizers form the argmin)

Remark 0.5 (Global minimizers form the argmin). The global minimizers of f are exactly the elements of $\arg\min_{x \in A} f(x)$ (Definition 0.14). A function can have more than one global minimizer. For example, $f(x) = (x^2 - 1)^2$ on $\mathbb{R}$ satisfies $f(x) \geq 0$ for every x, and $f(x) = 0$ exactly when $x = -1$ or $x = 1$, so its global minimizers are -1 and 1, and $\arg\min_{x \in \mathbb{R}} f(x) = \{-1, 1\}$.

Definition 0.17 (Local minimizer)**.** Let $A \subseteq \mathbb{R}^p$ and let $f : A \to \mathbb{R}$ be a function. A point $x^* \in A$ is a **local minimizer** of f if there is a number $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in A$ with $\|x - x^*\| < \delta$, where $\|\cdot\|$ is the Euclidean norm.

Remark (Local and global minimizers)

Remark 0.6 (Local and global minimizers). For $p = 1$, $\|x - x^*\| = |x - x^*|$. For example, with $x^* = 1$ and $\delta = 1$, the condition $|x - 1| < 1$ means $0 < x < 2$.
Every global minimizer (Definition 0.16) is a local minimizer: take any $\delta > 0$. For example, $x^* = 0$ is a global minimizer of $f(x) = x^2$ on $\mathbb{R}$, so it is also a local minimizer.
The converse fails. In Exercise 0.2, $x^* = 1$ is a local minimizer of $f(x) = x^3 - 3x$ (take $\delta = 1$), but not a global one.

Example 0.9 (A point that is not a local minimizer)**.** For $f(x) = x^3 - 3x$, the point $x = -1$ is not a local minimizer. For any t,

$$\begin{aligned}
f(-1+t) &= (-1+t)^3 - 3(-1+t) && \text{(substitute)}\\
&= (-1 + 3t - 3t^2 + t^3) - 3(-1+t) && \text{(expand the cube)}\\
&= -1 + 3t - 3t^2 + t^3 + 3 - 3t && \text{(multiply } -3 \text{ into the parentheses)}\\
&= 2 - 3t^2 + t^3 && (3t - 3t = 0 \text{ and } -1 + 3 = 2)\\
&= 2 + t^2(t-3) && \text{(factor out } t^2)\\
&= f(-1) + t^2(t-3), && (f(-1) = (-1)^3 - 3(-1), \text{ which equals } 2)
\end{aligned}$$

and $t^2(t-3) < 0$ whenever $t \neq 0$ and $t < 3$. So whatever $\delta > 0$ is, the point $x = -1 + t$ with $t = \min\{\delta/2, 1\}$ satisfies $|x - (-1)| < \delta$ and $f(x) < f(-1)$. For example,

$$\begin{aligned}
f(-0.9) &= -0.729 + 2.7\\
&= 1.971\\
&< 2.
\end{aligned}$$